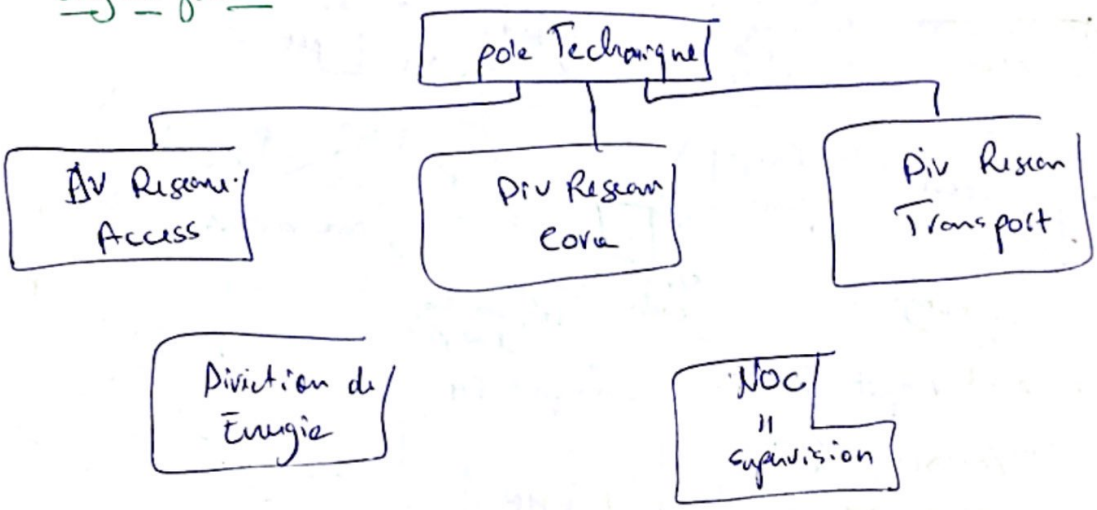
• Seminaire :

opérateur : entité qui fournit des services de communication aux personnels. pour mettent des infrastructures, services (téléphonie, accès aux internet, TV...) => les obérateurs interconnect for global Network.

Organigramme :



TDM :

commutation circuit => Time Division ~~Modulation~~ Multiplexing.

=> limitation aux Bande passante.

↳ Migration vers :

NGN : New generation network.

architecture basé sur les IP protocols. => packet commutation.

=> efficace, amélioré la bande passante.

migration using passerelles. (voip)

les réseau sont mis a niveau pour prendre les services NGN.

=> pour adapter sur la partie accès aux réseau.

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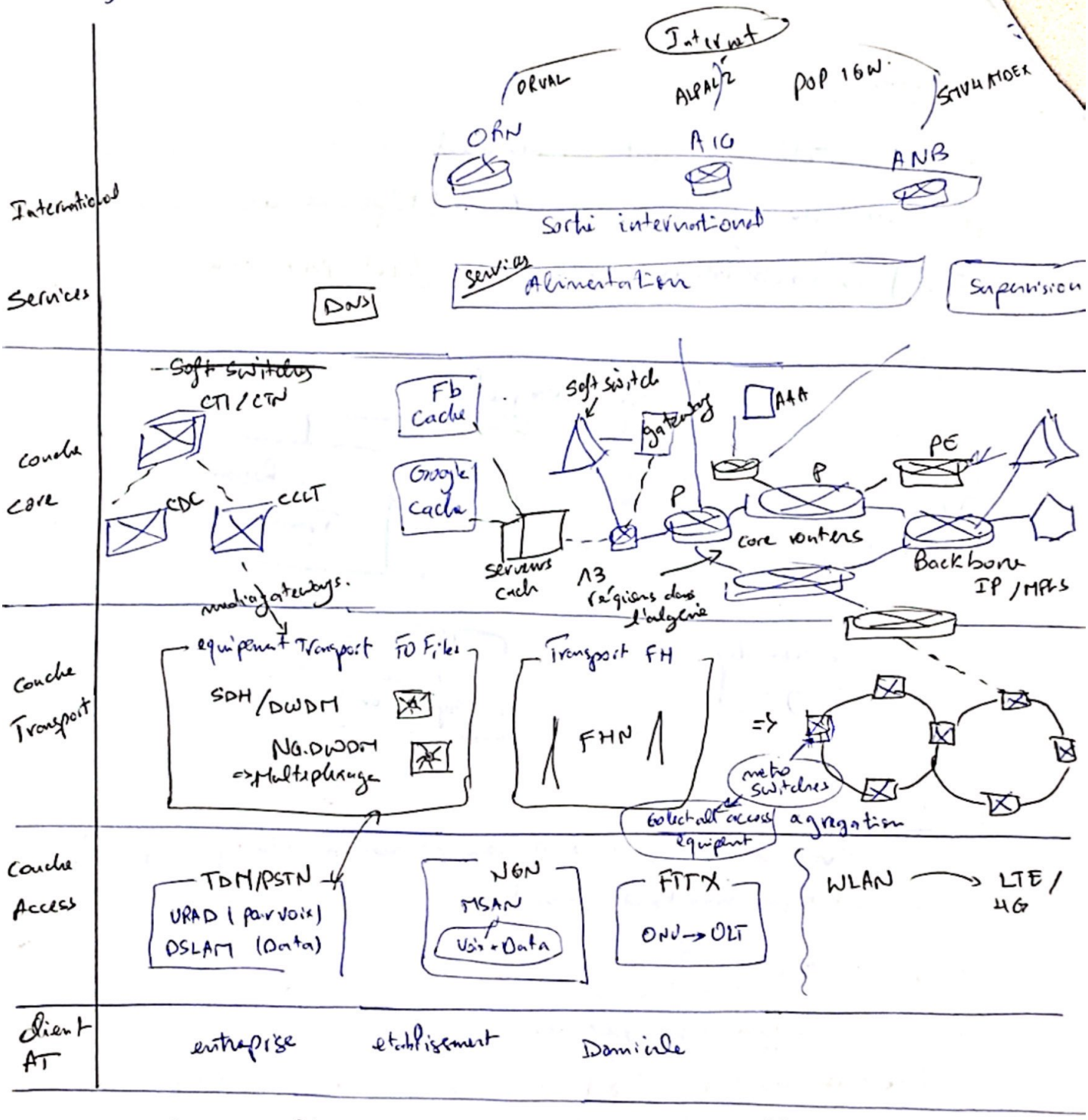
L'infrastructure des couches :



- each couche consider the couche beneath as its client

@ couche Access; core & content, transport (National, International)

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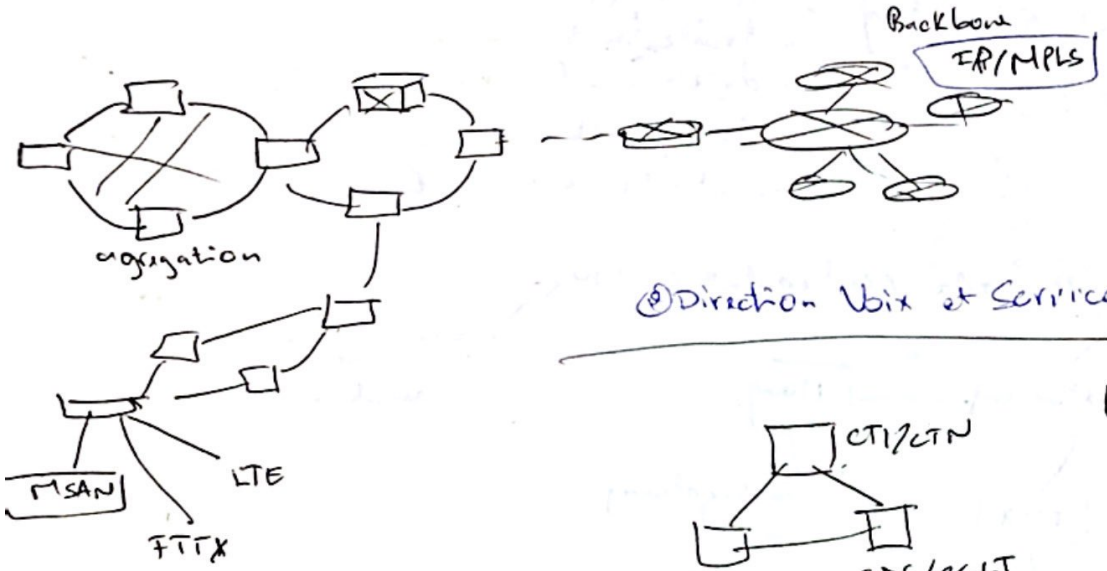
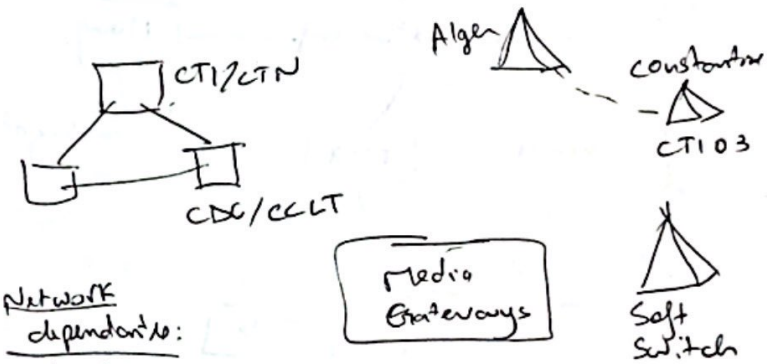
...vision Réseau core:

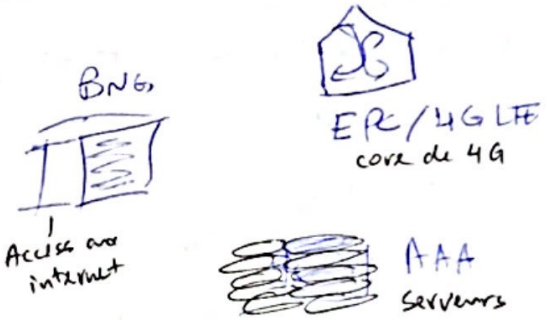
① Agregation:



② Direction Voix et Services:



③ Direction Plat forme:

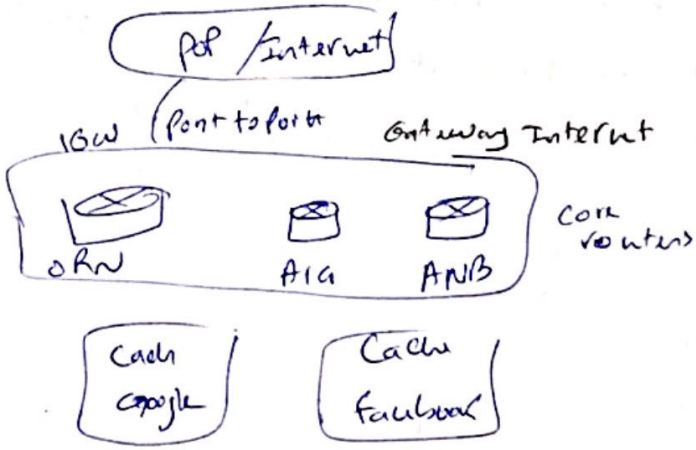


- Network dependantes:
  - NGN/TDM
  - IMS platforme
  - centre de transmit national et international

Réseaux Dependantes

- BNG: 18 noeuds national
- IPC core 4G
- AAA platforme

④ Direction Block Bone Internet



other cach servers.
⋮

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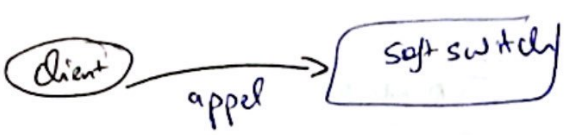
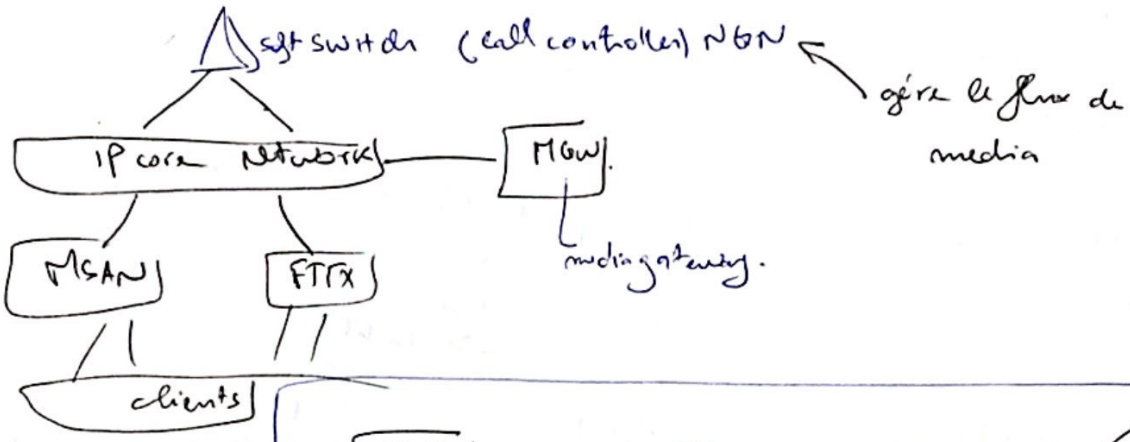
Soft Switch : réseau core qui est dédié à la voix acheminant et controllant les trafic des MSAN

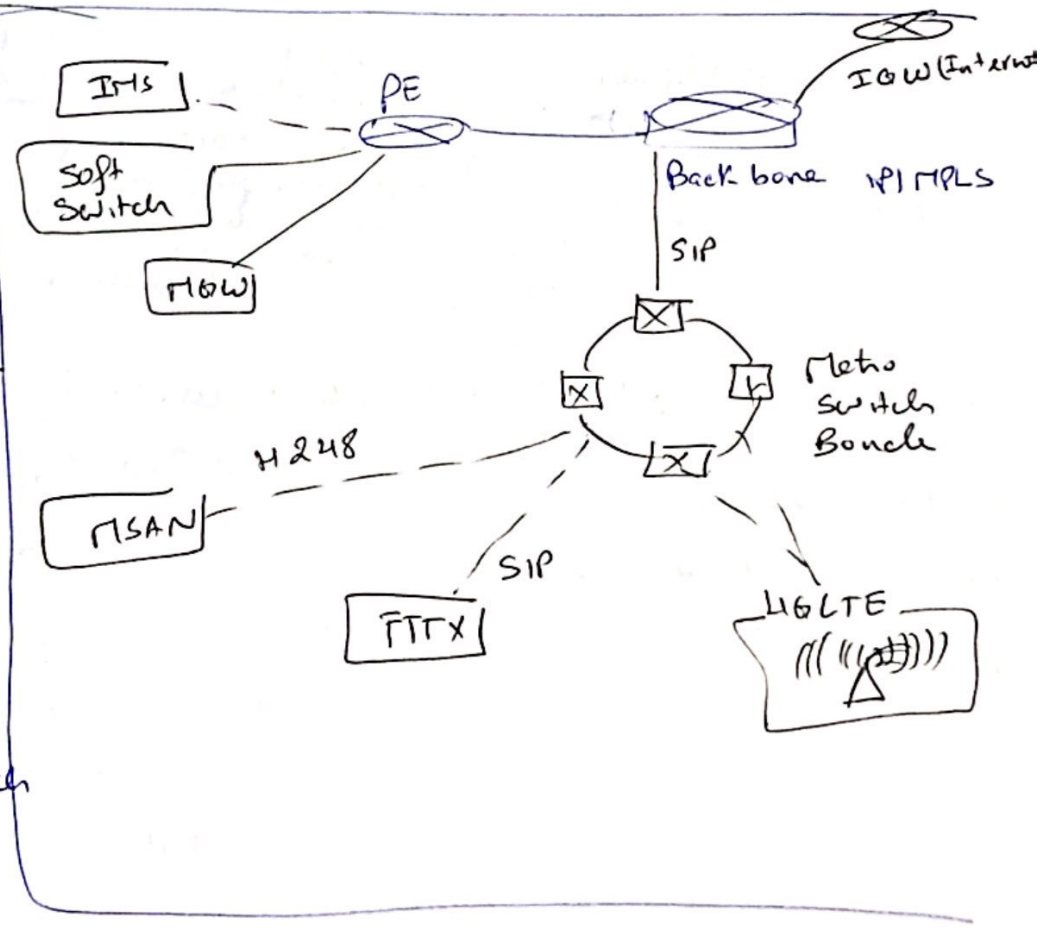


- reçoit l'appel
- traite la signalisation
- determine la disponibilité
- identification.

→ etablit la connection.



IMS : Réseau core dédié aux à la voix acheminant et controlle le trafic national des réseaux access FTTx et VoLTE



IMS : basé sur IP, fournit des services multimedias (voix, video ...) via réseaux IP

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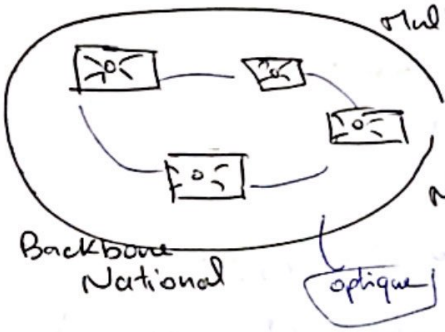
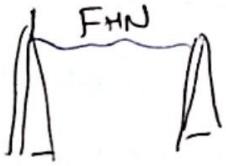
Division Transport :

① Direction Developpement Reseau Transport (partie Actif) = equipement.

→ Reseau National



- reseau FH :

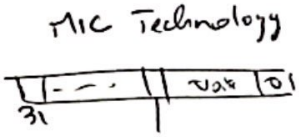


Backbone FH

Backbone : allerz et retoure

- actif = equipement qui ~~consome~~ necessite energy
- passif = " qui ~~consome~~ necessite pas

MIC Technology → PDH → SDH → OTN → DWDM - Dense Wave



PDH :
- 140 Mb
- synch Problem

SDH :
- sync fixed
- 10 GB debit

OTN :
- WDM → wavelength

DWDM : CDWM : course wave

NG-DWDM : Tera bits debit, grandes distances → fibre optique

3,15 Tb/s → 58 Milliards.

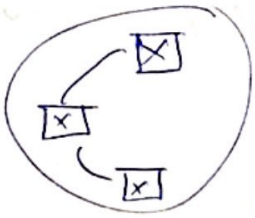
- reseau maillé 24 900 Km FOptique.
- FH → capacité 1GB → 10 GB in the way.

SDH 10/40 GB still used in some areas.

λ → Bande ⇒ ex : 400 GB par longeur d'onde dans NG-DWDM

↳ 32 longeur d'onde ..

② Direction Reseau Transport Internationales

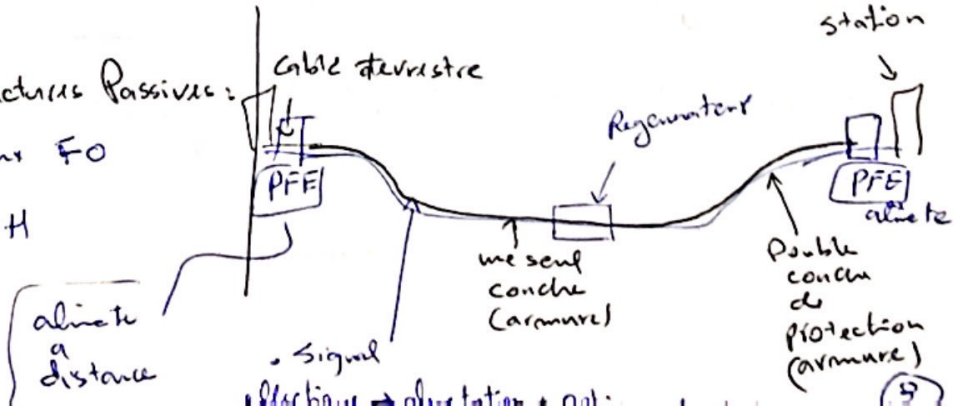


- Liaison terrestre avec Tunisie.
- Sous-marin transport < Travaux de cable / maintenance de cable

③ Direction Deverp Infrastructures Passives :

- Genie civiles Travaux FO
- Support Transport FH

↳ Deploiement des Réseaux.



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④ Direction Exploit et Maintenance Réseau Transport

~~MLMA~~ : maintenance de FO / FH.

- maintenance before en panne et après

Division Reseau Access :

lien entre clients et Services de telecom via un equipement

(DSLAM, MSAN, OLT et/ou BTS)

Reseau access filaire : Tx physique (cuivre CDN, fibre ODN)

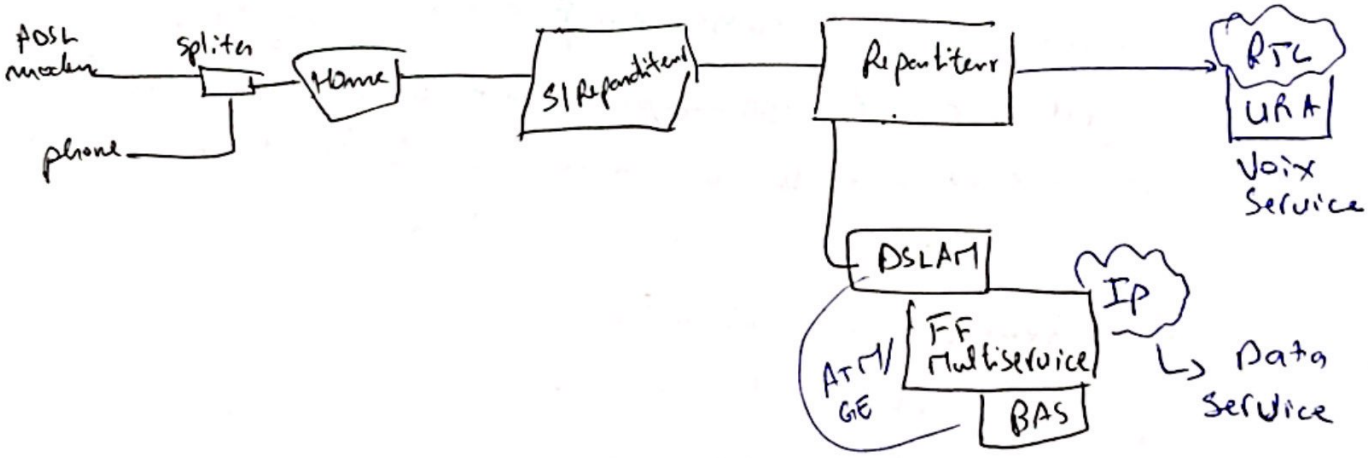
FTTx : x = lieu de terminaison (Curb, Building, Home) ...

- Topology : ① point to point

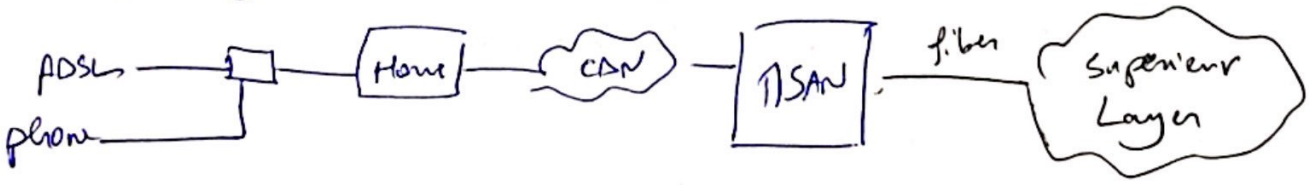
② point to multipoint :

Services :

① Using DSLAM (old, TDM and Multeplexing)



② Using MSAN : Multiservice (Voix and Data in one device)



MSAN : point-point, Multiservice, cuivre (CDN) Network

└ copper Distribution Network

SR armoir | jonctions cables => manchons | réglettes = bouclant client

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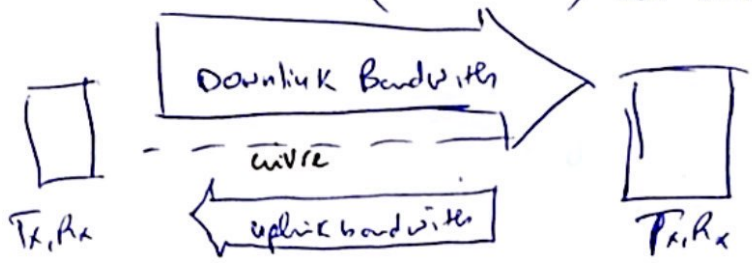
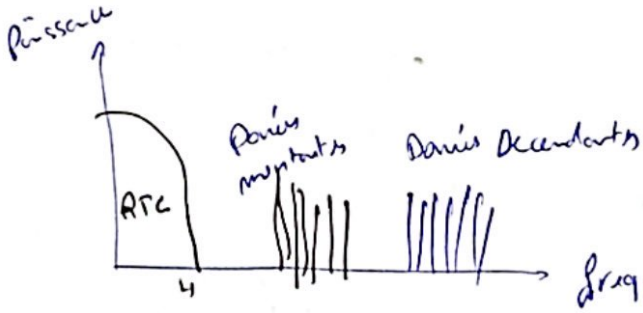
DSL Technology: methode de Tx haut debit (Voix et Data) sur Cuivre Lines

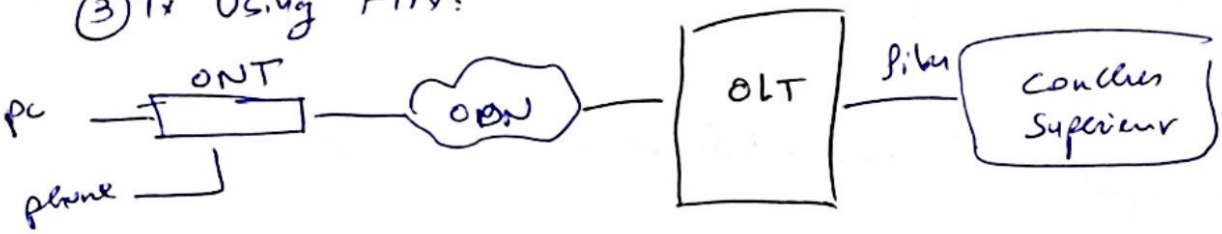
modulations et freq bands pour séparer les données des signaux montant et descendant



DSL
- ADSL
- VDSL
- DSL symetric



③ Tx Using FTTx:



OLT: optical Line Terminal: point – multip en fibre optique
- ODN: Optical Distribution Network
- converge services Voix et Data in one single Node

→ ODN: Network in fibers, components: SRO (repentiteur)
PBO (point de Branchement)
BPE (protection abonitiers)

GPON: adopte WDM to tx different wavelengths of Data dans le même ODN.

1290 — 1330 nm — amont (uplink) 1480 – 1500 nm — aval (Downlink)

reaches 2.5 GBits Debit pour 16 clients Downlink
1 GBits Uplink

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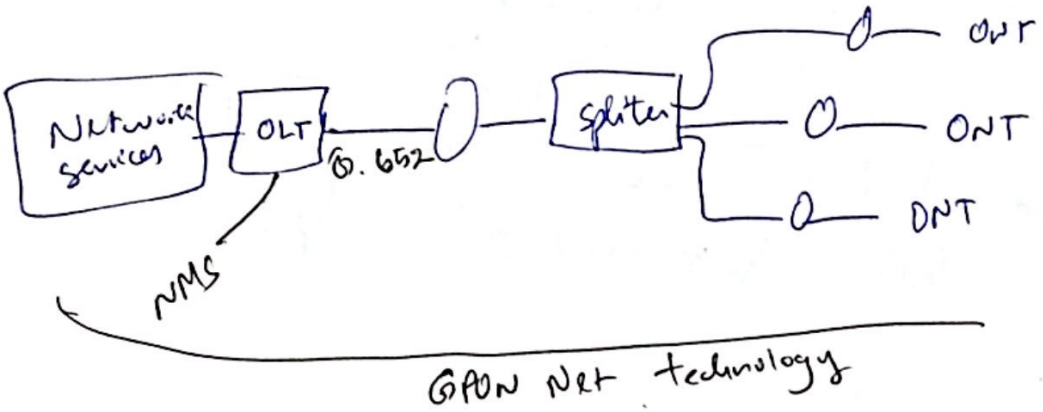
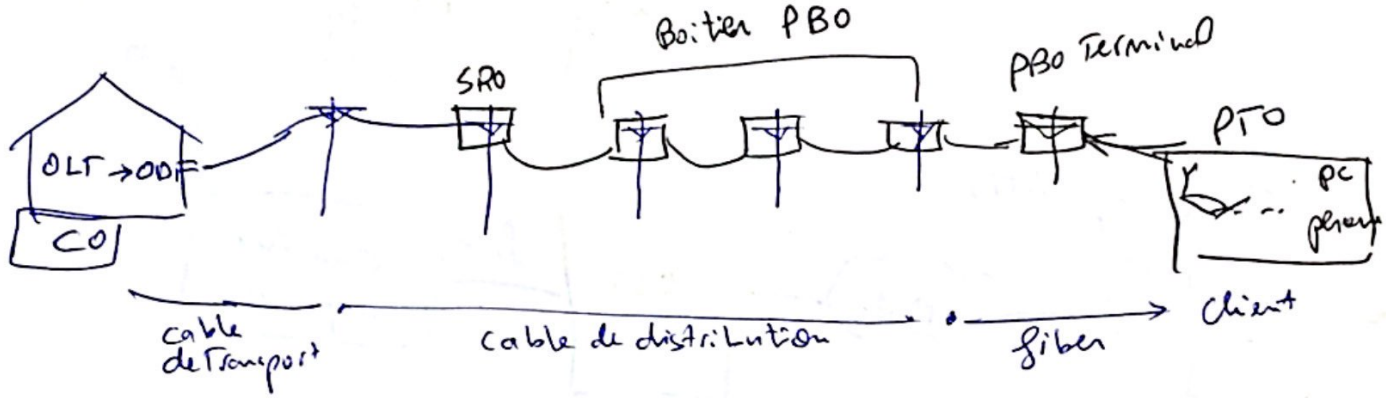


OPN solutions:



→ Technology GPON Point closing POW receivers...

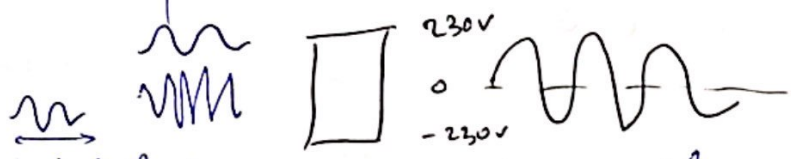
• Division Energy:

→ energy, refroidissement, redondance, securité

equipents:
- Transformateur de puissance
- cellules moyenne tension
- groupe electrogène
- Inverseur de source (Automatic Transfer Switch)
- distribution electrique.
- Uninterruptible powersupply:

pour alimenter

pour climatisation des data centers.



Neuf Défauts → → Reseau stable

- centrale de climatisation. (condenseur ...)

- climatisation précise.

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